

Supplementary Materials

Agreement Is Not Validity: Cross-Model LLM Consensus in Diagnosing Student Failure Modes in K-12 Math Tutoring Dialogue

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This document provides a detailed procedural accounting of the study reported in the main manuscript: the corpus and data-processing pipeline (Section 1), the failure-mode codebook and its development (Section 2), reference-set construction and sampling (Section 3), human annotation and expert-reference-set construction (Section 4), the LLM prompt design and implementation (Section 5), model configuration (Section 6), the formal definitions and implementation details of every evaluation metric (Section 7), and ethics/data governance (Section 8). All content below describes the pipeline and analyses underlying the reported results only; earlier study designs that were superseded prior to the current manuscript (e.g., a context-window-size ablation) are not described here, as they do not support any result reported in the main text. Identifying details about the data provider and platform are redacted ([anonymized]) consistent with the main manuscript’s double-blind review requirements.

S1 Data Provenance and Processing Pipeline

S1.1 Source Corpus

The source corpus is a large, de-identified collection of one-on-one K-12 mathematics tutoring transcripts provided by [anonymized], comprising 22,821 transcribed tutoring sessions and 35,072 individually graded student responses tied to specific curriculum learning objectives, of which 10,435 responses were graded as incorrect by the platform’s deterministic grading process based on the students’ submitted responses in the on-screen answer fields. All identifying information (student and tutor names, school identifiers) had already been removed by the data provider prior to our access.

S1.2 Working Sample and Localization

From the 10,435 platform-graded incorrect responses, 5,000 were sampled without replacement (capped at 2 responses per session to limit session-level clustering; random seed 20260720), spanning 4,749 unique sessions. For every sampled response, the surrounding dialogue was automatically *localized*: an LLM-based localization pass identified, within each session, the tutor’s setup question (Q), the student’s graded response (R), and—where present—the tutor’s subsequent confirmation or correction (C), for every learning-objective segment in the session (8,005 localizations total across the 4,749 sessions, including learning objectives not among the 5,000 sampled responses but needed for correct segmentation). Each localization carried a model-reported confidence rating; restricted to the 5,000 sampled responses, the confidence distribution was 74.9% high, 24.1% medium, and 1.0% low. Localization coverage was independently spot-checked on a random sample of 50 sessions against the full set of learning objectives recorded in the source metadata, with zero coverage gaps found.

S1.3 Data-Quality Filtering

Two additional data-quality issues were identified and screened for prior to any failure-mode annotation:

- **Speaker role-swap.** A source-data transcription artifact was discovered in which a session’s tutor and student turns were occasionally mislabeled (i.e., diarization assigned an utterance to the wrong speaker). This was verified as a data artifact, not a pipeline error, by direct inspection of the raw transcripts, and confirmed with the data provider. An automated session-level detector flagged 536 of the 4,749 sessions (11.3%) as affected; this is a documented floor estimate, not a full-prevalence estimate, since manual spot review found the automated detector under-catches some cases. A second, targeted item-level check (applied specifically to the graded response R) additionally flagged 656 of the 5,000 sampled responses.
- **Localization reliability.** Items with low localization confidence, or with an invalid/out-of-range turn assignment, were flagged separately.

The final clean pool excludes the union of: low-confidence localizations (49 items), role-swap-flagged sessions (559 items), the targeted role-swap check (656 items), and invalid turn ranges (29 items). These categories overlap (e.g., a role-swap-flagged session may also have a low-confidence localization), so the counts are not additive; in total, 1,146 of the 5,000 sampled responses (22.9%) were excluded, leaving **3,854 clean error moments (77.1%)**—the pool referenced in the main text as the starting point for reference-set construction (Section 3).

Table 1: Derivation of the 3,854-item clean pool from the 5,000-item working sample.

Stage	Count
Platform-graded incorrect responses (source corpus)	10,435
Sampled working set (2/session cap)	5,000
Excluded: low-confidence localization	−49
Excluded: role-swap-flagged session	−559
Excluded: targeted role-swap check (item-level)	−656
Excluded: invalid turn range	−29
<i>(categories overlap; union excluded, not summed)</i>	−1,146 total
Clean pool	3,854

S2 Failure-Mode Codebook

S2.1 Origin and Iterative Refinement

The five-class failure-mode taxonomy (Table 1 of the main text) originates in [anonymized]’s own operational diagnostic codebook, already in use to characterize incorrect student responses at deployment scale. We formalized this codebook into an explicit, ordered decision procedure and LLM prompt, then iteratively refined both the decision procedure and the accompanying worked examples over approximately ten rounds of revision. Each round was motivated by systematic review of concrete disagreement cases between LLM predictions and expert human judgment—never by hypothetical edge cases—and each revision was subsequently evaluated on a freshly drawn, disjoint holdout rather than re-checked against the same items that motivated it, following standard human-in-the-loop, error-analysis-driven prompt engineering practice. This iterative process also surfaced a structural revision to the taxonomy itself: the original single *Unknown* (UNK) catchall

class was split into two conceptually distinct labels—*Uncertainty* (UNC), reserved for cases where the student’s own graded response expresses genuine hedging or reports a technical issue, and *Not Applicable* (NA), an exclusion flag for cases where the classifier itself (human or LLM) cannot confidently assign any of the five substantive failure modes, or where the item is not a genuine failure-mode-diagnosis case at all (e.g., a data artifact, an off-task remark, or a response that appears correct despite being marked incorrect). This UNC/NA split is used throughout the reference-set construction, annotation, and LLM classification described below.

S2.2 Key Revisions Reflected in the Final Codebook

Relative to an earlier, less-refined version of the decision procedure used only during intermediate prompt-development cycles, the final codebook (reproduced in full in Section 2.3) incorporates four revisions, each motivated by a specific, recurring disagreement pattern found via error analysis:

1. **Anchoring to the graded response.** The uncertainty (rule 1) and misattribution (rule 2) triggers are explicitly restricted to the student’s graded response (R) itself, rather than any hedge or off-problem reference appearing elsewhere in the dialogue window. Earlier prompt versions allowed uncertainty expressed at an earlier turn to override a later, definite incorrect response.
2. **The Procedural Slip / Conceptual Gap “mirror-image” rule.** A response is not automatically classified as a Procedural Slip merely because the overall strategy sounds reasonable; the codebook now explicitly instructs checking whether the specific step that went wrong reveals a missing sub-skill (e.g., building an equivalent fraction, place value, multi-digit comparison), which indicates a Conceptual Gap instead. This was the single largest confusion pattern identified in earlier error analysis.
3. **The bare-answer rule.** A hedged or question-marked bare numeric answer with no supporting reasoning is classified as Uncertainty; a flat, confident declarative bare answer with no reasoning defaults to Conceptual Gap rather than the Not-Applicable catchall.
4. **Deriving the expected answer from context.** The classifier is instructed to work out the expected answer or approach from the problem information already present in the dialogue window when possible, rather than treating the absence of an explicitly stated correct answer as grounds for a Not-Applicable classification.

S2.3 Final Codebook Text

The following reproduces the operational failure-mode codebook used for LLM classification, with minor clarifications to terminology and analysis treatment that do not alter the classification decision rules.

You are diagnosing the type of mathematical failure mode behind a student’s incorrect answer in a one-on-one tutoring session.

You will see: the learning objective being assessed, and a window of conversation turns leading up to and including the student’s graded response (R). The response has already been confirmed incorrect -- your job is to diagnose WHY, using this ordered decision procedure. Check the conditions IN ORDER and stop at the first one that applies -- do not skip ahead, and do not pick a later class just because it also seems to fit:

1. If the student’s response AT THE GRADED TURN (R) ITSELF expresses uncertainty (e.g. "I don’t know," "wait, what?", or a trailing hedge/question mark on an otherwise bare answer

like "Is it 28?" or "550,000?") or reports a technical problem (audio, connection) -> UNC. An earlier hedge elsewhere in the window does NOT qualify unless R itself also shows it -- a student who hesitates mid-work but goes on to give a definite, specific final answer at R should be diagnosed from that final answer, not the earlier hesitation. IMPORTANT: a hedge opener ("I think," "wait," etc.) does NOT automatically trigger UNC if the response goes on to commit to something specific -- only fall back to UNC when the response doesn't offer substantive follow-through. Example: "I think they're both correct because 0.07 is the same as 7 tenths..." -- the "I think" is a conversational softener, not genuine uncertainty; diagnose the underlying fraction/decimal failure mode (CON) instead. Contrast with a bare "I think it's 24" with nothing else, which stays UNC-eligible since there's no substantive follow-through. Also: a response that clearly hedges and then cuts off (e.g. "Grouping is like when you like, um, like have the--") is still UNC, not NA -- NA's "cut off with no content" trigger (see rule 6 below) means nothing interpretable was said at all, not the same as a hedge that already fired before trailing off. IMPORTANT: if the uncertainty co-occurs with numbers/values pulled from a different part of the problem than the one being asked, check CTX (rule 2) first -- that's CTX, not UNC.

2. If R itself uses numbers or context from a DIFFERENT question/problem than the one being asked -> CTX. This trigger must come from what R itself shows -- not a different sub-question discussed elsewhere in the excerpt. This includes explicitly saying the answer was recalled or guessed from a different, similar problem rather than worked out for this one (e.g. "There was a question like this in my book, so I just remembered the answer") -- CTX applies even if the specific wrong value itself doesn't obviously look out of place. IMPORTANT: this requires a genuinely DIFFERENT problem -- substituting a transposed or misheard digit from the SAME problem's own given numbers (e.g. using 190 instead of the given 119) is NOT CTX, it's a same-problem execution/reading slip (PROC or CON depending on what else the response shows). ALSO IMPORTANT: a large numeric miss alone is NOT evidence of CTX -- check whether the number is traceable to the current problem's own given inputs (even via a wrong operation), not how far off it is from the correct answer. If traceable -- the current problem's own numbers, just combined or shifted incorrectly -- that's CON/COMP/PROC territory regardless of magnitude, not CTX. Example: " $4.82 \times 10 = 40,820$ " is off by $\sim 1000\times$ from the correct 48.2, but 4.82 is the current problem's own number -- that's CON (a garbled place-value misunderstanding), not CTX, despite the large miss.

3. If the student chooses the wrong TYPE of operation for this specific question (e.g. adds when the problem calls for multiplication) -> COMP. IMPORTANT: this requires a wrong operation TYPE, not just a wrong parameter within the right type -- choosing the wrong number to divide/multiply/add/subtract by, while still correctly picking division/multiplication/addition/subtraction as the general operation, is NOT COMP. That's a wrong idea about what number to use, which is CON (check 4). Example: dividing by 2 instead of 4 to find 25% of a number -- division is the right operation type, so the error is CON (misunderstanding what 25% means), not COMP.

4. If their reasoning reveals a wrong underlying mathematical idea (e.g. believes $0 \times n = n$; multiplies exponents instead of adding them) -> CON

5. Otherwise, if the task type/strategy is correct but the calculation or execution is wrong (e.g. correct method, arithmetic slip) -> PROC

Precedence examples:

- A response that could plausibly read as BOTH "wrong operation" (COMP, check 3) and "wrong underlying idea" (CON, check 4) is COMP, not CON, because COMP is checked first.
- Same logic for CTX vs. CON: if the response pulls in numbers/methods from a different question, that's CTX (check 2), not CON (check 4), even if it also reveals faulty reasoning -- CTX is checked first.

The mirror-image trap -- don't default to PROC just because the overall strategy sounds reasonable: check whether the SPECIFIC step that went wrong reveals the student doesn't understand a sub-skill needed for that strategy (building an equivalent fraction, finding a common multiple, aligning place value, comparing multi-digit numbers). If so, that's CON, even though the general approach was fine. PROC requires the method AND reasoning to be demonstrably intact, with only a value miscounted, mis-copied, or a small arithmetic slip -- not a wrong sub-skill application. Example: a student converts $2/3$ to " $6/4$ " instead of " $4/6$ "

as an equivalent fraction -- the overall plan (convert, then subtract) is right, but building the equivalent fraction itself is wrong: that's CON, not PROC. Decimal-point-side confusion is a specific instance of this trap: writing a digit on the wrong side of the decimal point, or inserting a spurious zero when shifting by a power of ten, is a missing sub-skill in decimal notation, not a one-off slip. Example: "40.82" instead of "48.2" for 4.82×10 , or "4.0" instead of "0.4" for $4/10$ -- CON, not PROC, even though the earlier steps (reaching "4/10," multiplying by the right power of ten) were correct.

IMPORTANT -- weigh the tutor's own words as direct evidence, not just the size of the numeric gap: if the tutor's own next turn (C) explicitly confirms the approach/process itself was correct and only flags the final answer or a specific step, that's strong evidence for PROC -- even if the resulting error looks conceptual at a glance. Example: tutor says "your approach was good... I just wanted you to double-check your final answer" right before pointing out a basic multiplication-fact slip (4×3 stated as 14 instead of 12) mid-procedure -- PROC, not CON, on the strength of the tutor's own confirmation.

Bare wrong answers (just a final number/phrase, with no work or explanation shown): this is exactly where rule 1's hedge/question-mark trigger above matters most, since there's often no other content to go on. A hedge or trailing question mark attached to a number that reveals nothing specific ("Is it 28?", "550,000?") IS expressing uncertainty per rule 1 -> UNC. BUT if the hedged number is itself diagnostically specific -- clearly revealing a particular wrong operation or idea -- diagnose from that number instead, same principle as rule 1's hedge-opener exception. Example: "0.18?" for " 1.8×100 " -- the hedge doesn't erase that 0.18 clearly reflects dividing by 10 instead of multiplying by 100 (CON), not just an uncertain guess. A flat, confident declarative bare answer with no hedge ("12.", "2/3.") does NOT meet rule 1's trigger -> default to CON under this operational codebook rather than NA -- unless truly uninterpretable (inaudible, cut off with no content at all), which remains NA via the catchall (check 6).

When possible, work out what the correct answer or approach should be from the problem information already in the window (the learning objective and preceding turns), and use that to judge whether the student's error is small/execution-level (favors PROC) or reflects a substantially different value or approach (favors CON/COMP). Don't treat "the correct answer isn't explicitly stated" as grounds for the NA catchall if you can reasonably work it out yourself from what's given.

6. NA (catchall -- NOT a failure mode type, an exclusion flag): if you are genuinely uncertain which class applies even after working through steps 1-5 above, your primary_label must be NA -- not a low-confidence guess at UNC/CTX/COMP/CON/PROC. This is separate from check 1 (the STUDENT expressing uncertainty at R, which is UNC); NA is about YOUR OWN inability to classify, OR about the item not being a genuine failure-mode-diagnosis case at all. Six specific, recurring triggers for NA (not the only ones, but the most common): the response is just "[unclear]" or otherwise leaves nothing interpretable to go on (inaudible, cut off with no content at all -- not to be confused with a hedge, which is check 1); the response is a bare acknowledgment ("Yeah," "Okay," "Mm-hmm") with no independent content -- BUT only when there's nothing specific enough to diagnose from either the student's words or what they're agreeing to; if the tutor's immediately preceding turn proposed something SPECIFIC (a particular operation, value, or approach) and the student's agreement endorses that specific thing, diagnose normally from what was endorsed instead (e.g. agreeing "should we multiply here?" when division was needed is still COMP, not NA -- the endorsement itself is diagnostic content); R ITSELF appears to be mislabeled by a tutor/student role swap -- you can't tell whether R was actually said by the student or the tutor (a known transcript data artifact). If a swap happens elsewhere in the excerpt but R itself is clearly and correctly attributed, that is NOT grounds for NA -- diagnose R normally, same anchor-to-R principle as checks 1 and 2; the response actually appears correct despite being marked incorrect (a known residual label-noise issue -- don't force a failure mode diagnosis onto an answer that looks right to you; NOTE: this is about the final answer being right with nothing else to diagnose -- it does NOT cover a right answer reached via flawed reasoning. If the explanation itself reveals a real problem even though the terminal number matches,

diagnose that reasoning normally, most likely CON or PROC -- don't default to NA just because the number is right. Explicit tutor praise or confirmation in the very next turn (C), e.g. "excellent job," "well done," "that's correct," is strong evidence for this trigger when C is visible -- check what the tutor's own words actually say rather than assuming the item was correctly marked incorrect); the response is an opinion, reflection, or off-task remark rather than an attempt at the math problem itself (e.g. commenting on the lesson, a personal aside) -- there's no math reasoning to diagnose, so it isn't a failure mode at all; or R just reads back or restates the question itself, with no actual attempt to solve it -- no answer, no work, no reasoning offered beyond repeating what was asked (a known gap this closes: "student reading the question back" was always meant to be filtered upstream in Phase 1a-ii, but the mechanical filter can't reliably detect it -- this NA trigger catches what slips through). NA is not treated as a sixth substantive failure mode, but it is retained as a classification outcome in the reported evaluation so that appropriate abstention from diagnosis can be assessed alongside the five substantive failure-mode labels.

Note: some conversation windows are short (e.g. just the question and response) and won't show much reasoning -- diagnose from what's there; a short window is not itself evidence of NA or UNC.

S3 Reference-Set Construction and Sampling

S3.1 The 330-Item Refinement Pool

Of the 3,854-item clean pool, 330 items were drawn and stratified by a preliminary, provisional LLM-assigned label into three disjoint sub-pools used exclusively for codebook and prompt development: a 250-item annotation-development sample, a 40-item few-shot candidate pool, and a 40-item calibration candidate pool. These 330 items were used throughout the iterative refinement process described in Section 2 and were never used for the reference set analyzed in the main text; disjointness was verified programmatically (zero overlap) before the reference set was drawn.

S3.2 Reference-Set Sampling ($n = 100$)

The 100-item reference set was drawn from the remaining 3,524 items (3,854 minus the 330-item refinement pool), disjoint from all three refinement sub-pools by construction. Sampling (random seed 20260810) used a fixed, disproportionate stratified allocation by preliminary (provisional) LLM-assigned label—not a random or natural-proportion draw—deliberately over-weighting the classes that the preceding error-analysis process had identified as the most confusable: *Conceptual Gap*, *Procedural Slip*, and *Operator Selection*. *Misattribution* (CTX) was capped at 10 because only 20 unused CTX items remained in the entire 3,854-item pool after the refinement-pool draws. Table 2 gives the exact allocation. Because the reference set was designed to stress-test classification at difficult class boundaries rather than to estimate population prevalence, this provisional-label allocation should not be interpreted as reflecting the natural class distribution in the corpus; the final human-adjudicated label distribution (Section 4) differs substantially from it, since many provisionally-labeled Procedural Slip and Operator Selection items were reclassified—predominantly to Conceptual Gap or Not Applicable—once expert annotation was complete.

S3.3 Dialogue Context per Item

Unlike an earlier study design (superseded prior to the current manuscript) that manipulated dialogue-window length as an experimental condition, the reference set analyzed in the main text uses complete localized learning-objective segments, yielding naturally varying context lengths

Table 2: Reference-set sampling allocation by preliminary (provisional) LLM-assigned label, $n = 100$.

Preliminary label	Items sampled
Conceptual Gap (CON)	28
Procedural Slip (PROC)	24
Operator Selection (COMP)	20
Uncertainty (provisional UNK)	18
Misattribution (CTX)	10
Total	100

across items (2–133 turns; $M = 7.7$). For the reported LLM classifications, each excerpt was truncated at the student’s graded response (R), excluding any subsequent tutor confirmation or correction (C) to avoid lookahead bias.

S4 Human Annotation and Expert Reference-Set Construction

Two trained researchers (Rater 1 and Rater 2) independently applied the final codebook (Section 2.3) to all 100 reference-set items, each blind to the other’s labels. For every item, each rater recorded a primary failure-mode label, a confidence rating (low/medium/high), and a written justification.

S4.1 Initial Agreement

Independent raw agreement between Rater 1 and Rater 2 was 71/100 (71%; Cohen’s $\kappa = .588$, 95% CI [.454, .707], computed via the bootstrap procedure described in Section 7).

S4.2 Consensus Adjudication

The 29 items on which the raters initially disagreed were resolved through structured consensus review: both researchers jointly reconsidered their original labels and justifications against the source transcript, using a dedicated adjudication tool that displayed both raters’ original label, confidence, and justification side by side, until reaching a single agreed-upon label, confidence rating, and consensus justification for each item. Of these 29 adjudicated items, the consensus label matched Rater 1’s original label in 22 cases, matched Rater 2’s original label in 6 cases, and matched neither rater’s original label in 1 case (a response judged, on joint reconsideration, to contain insufficient information to support any of the five substantive failure-mode diagnoses, and therefore assigned *Not Applicable*).

S4.3 Final Expert Reference Set

The resulting 100-item expert reference set—71 items on which the raters independently agreed, plus 29 consensus-adjudicated items—is the single human reference against which all LLM classifications are compared in the main text. Table 3 gives its final label distribution.

Table 3: Final consensus expert reference-set label distribution ($n = 100$).

Label	Count
Conceptual Gap (CON)	45
Not Applicable (NA)	21
Uncertainty (UNC)	14
Misattribution (CTX)	8
Procedural Slip (PROC)	6
Operator Selection (COMP)	6
Total	100

S5 LLM Prompts

Every LLM classification reported in the main text used the identical prompt-construction logic described here, regardless of vendor, to ensure that any difference in results reflects the model rather than prompt drift between conditions.

S5.1 System Prompt

The system prompt concatenates (1) the final codebook text in full (Section 2.3), followed by (2) a fixed, ordered sequence of worked few-shot examples (Section 5.3), each formatted as a tagged conversation excerpt followed by its correct diagnosis, and closing with an instruction to diagnose the upcoming new item using the same procedure. The exact assembly logic (Python) was:

```
def build_system_prompt(fewshot_examples):
    blocks = [CODEBOOK_TEXT,
              "\nWorked examples (one per class, correct diagnoses shown):\n"]
    for ex in fewshot_examples:
        lines = [tag_turn(t, ex.q_turn, ex.r_turn, None) for t in ex.excerpt]
        blocks.append(
            f"---\nLEARNING OBJECTIVE: {ex.learning_objective}\n"
            f"CONVERSATION:\n" + "\n".join(lines) +
            f"\nDIAGNOSIS: {ex.primary_label}\n"
        )
    blocks.append(
        "---\nNow diagnose the following new item using the same procedure."
    )
    return "\n".join(blocks)
```

Each dialogue turn is inline-tagged to disambiguate the setup question, the graded response, and the tutor’s post-response turn:

```
[turn_number] role: content    <-- GRADED RESPONSE (R): diagnose THIS turn
[turn_number] role: content    <-- Q (setup question, context only)
[turn_number] role: content    <-- C (tutor’s post-response confirmation/
                                correction, context only)
```

S5.2 User Prompt

For each item to be classified, the user prompt supplied the learning objective, the full tagged conversation excerpt (Section 3.3), and an explicit instruction restricting the uncertainty and misattribution checks to the graded turn specifically:

LEARNING OBJECTIVE: {learning_objective}

CONVERSATION (Q = setup question, R = the graded response you must diagnose, C = tutor’s post-response turn if shown -- see inline tags below):
{tagged conversation turns}

Diagnose the failure mode in R (turn [r_turn]) using the ordered decision procedure. Checks 1 and 2 (UNC and CTX triggers) must be evaluated from R’s own tagged turn specifically, not Q or C.

For all classifications reported in the manuscript, C was omitted from the target item, consistent with the blinded condition described in Section 6.

S5.3 Few-Shot Examples

The system prompt embeds a fixed set of 17 worked few-shot examples—each a real, de-identified excerpt from the corpus paired with its correct expert-verified diagnosis—presented in a single fixed order, held constant across every classified item and every model (no per-item or per-model shuffling). Every example’s label, class-relative position, and content were included exactly as listed in Table 4; no worked example includes an explanatory rationale beyond the diagnosis label itself (i.e., the few-shot demonstrations are input–output pairs, not chain-of-thought traces). The set is disjoint from the 100-item reference set and was drawn from the 330-item refinement pool (Section 3).

Table 4: Composition of the 17-item few-shot set, in prompt order.

#	Label	Response ID
1	Operator Selection (COMP)	beboysp
2	Operator Selection (COMP)	bpuqfcj
3	Operator Selection (COMP)	ijazsam
4	Conceptual Gap (CON)	csacnwk
5	Conceptual Gap (CON)	dsjlnrt
6	Conceptual Gap (CON)	lqqsxht
7	Misattribution (CTX)	jhwchla
8	Misattribution (CTX)	lnwrqje
9	Misattribution (CTX)	gosrhbt
10	Procedural Slip (PROC)	fizewtd
11	Procedural Slip (PROC)	ixvdsdg
12	Procedural Slip (PROC)	ayltdkp
13	Uncertainty (UNC)	igzpsgg
14	Uncertainty (UNC)	mvxiyar

Table 4: Composition of the 17-item few-shot set, in prompt order.

#	Label	Response ID
15	Uncertainty (UNC)	duggyyr
16	Conceptual Gap (CON)	ioqhmat
17	Procedural Slip (PROC)	kmckpqd

The set was expanded from an earlier 15-example version (3 examples per class, balanced across the five substantive classes) to the 17-example version above by adding examples #16 and #17, which were added specifically to demonstrate the Conceptual-Gap/Procedural-Slip boundary described in Section 2 (item 2): #16 illustrates the sub-skill-gap pattern (a plausible-sounding but incorrect rule for adding fractions with unlike denominators), and #17 illustrates the boundary from the other side (a correct division and correct remainder, with the final answer failing only to account for the remainder—still a Procedural Slip, since the computation itself is correct). The three Uncertainty examples were originally labeled under the pre-split UNK catchall and are relabeled UNC at prompt-construction time, consistent with the taxonomy revision described in Section 2; their underlying content is unchanged. For illustration, one complete worked example (#1) is reproduced below exactly as it appears in the system prompt:

```

---
LEARNING OBJECTIVE: Writing simple fractions as decimals.
CONVERSATION:
[23] tutor: Okay. So [student], what should we do to write a fraction
as a decimal? What is our first step?
[24] student: We have to first figure out what it is as a
percentage, and then you can do it as a decimal. <-- GRADED
RESPONSE (R): diagnose THIS turn
[25] tutor: Mm-hmm. Yeah, so first we have to convert the
denominator to a 10 or 100 or a multiple of 10, 100, or 1,000.
All right? So we will be able to convert it to a decimal. OK?
DIAGNOSIS: COMP

```

S5.4 Structured Output Schema

All classifications used schema-constrained structured outputs restricted to the six allowable labels (UNC, CTX, COMP, CON, PROC, NA), a confidence rating, and a free-text justification. The canonical (vendor-neutral) schema was:

```

{
  "type": "object",
  "properties": {
    "primary_label": {"type": "string",
                      "enum": ["UNC", "CTX", "COMP", "CON", "PROC", "NA"]},
    "confidence": {"type": "string",
                   "enum": ["high", "medium", "low"]},
    "justification": {"type": "string"}
  },
  "required": ["primary_label", "confidence", "justification"],

```

```
"additionalProperties": false
}
```

Each vendor’s native structured-output mechanism was used to enforce this schema (Section 6).

S6 Model Configuration

S6.1 Models and Vendor-Specific Implementation

For comparability within this study, we use the terms *tier 2* and *tier 1* as study-specific labels for the selected cross-vendor capability levels rather than as vendor-defined universal categories. Three tier-2 models, one per vendor, were used for the primary analysis, plus one tier-1 Google model at maximum reasoning effort for the robustness check. Model selection was verified against each vendor’s current documentation at the time of execution. Table 5 summarizes the configuration; implementation differed by vendor because no single structured-output mechanism is shared across all three APIs.

Table 5: Model configuration by vendor.

Vendor	Model	Role	Structured-output mechanism
OpenAI	GPT-5.6 Terra	Tier-2 primary	Native <code>response_format</code> (JSON schema, strict mode)
Google	Gemini 3.5 Flash	Tier-2 primary	Native <code>response_schema</code> , with <code>response_mime_type</code> set to <code>application/json</code>
Anthropic	Claude Opus 5	Tier-2 primary	Forced tool-use (<code>tool_choice</code> fixed to a single schema-defined tool), the standard mechanism given the absence of a native structured-output equivalent
Google	Gemini 3.6 Flash	Tier-1 robustness check	Same as Gemini 3.5 Flash, with <code>thinking_budget</code> set to 65,535—the verified maximum value accepted by the API at the time of the study—as the closest available proxy for maximum reasoning effort

No explicit temperature or other sampling-parameter override was applied for any vendor; each model was called with its API default decoding configuration alongside the structured-output constraint.

S6.2 Blinded Classification Condition

Every classification reported in the main text used a single *blinded* condition: the model received the student’s graded response and all preceding dialogue in the localized segment, but not the tutor’s subsequent confirmation or correction (turn C), even when C was present in the transcript. This mirrors the information available to a real-time tutoring system, which must interpret a student’s response before any correction has been given, and avoids lookahead bias.

S6.3 Lookahead (With-Correction) Robustness Check

The blinded condition above creates a genuine information asymmetry relative to the human reference set: the expert raters (Section 4) annotated using the complete localized segment, including the tutor’s subsequent confirmation or correction (C) when present (Section 3.3), whereas the reported LLM classifications did not have access to C. We therefore used the blinded condition for the primary analysis because it reflects the information available to a tutor or AI tutoring system at the moment a student’s response must be interpreted: the preceding dialogue and student response, but not a subsequent correction that has not yet occurred. To assess whether this asymmetry, rather than a genuine reasoning gap, accounts for the observed human–LLM disagreement, we ran a supplementary *with-C* condition: identical to the blinded condition except that turn C was included in the excerpt whenever present, for all four model/tier configurations in Table 5. Table 6 reports Cohen’s κ against the expert reference set for both conditions.

Table 6: Blinded vs. with-C (lookahead) agreement with the expert reference set, by model.

Model	Blinded κ	With-C κ	Δ
GPT-5.6 Terra (tier 2)	.490	.461	−.029
Gemini 3.5 Flash (tier 2)	.574	.592	+.018
Claude Opus 5 (tier 2)	.524	.525	+.001
Gemini 3.6 Flash (tier 1, max reasoning)	.572	.553	−.019

Across all four configurations, agreement changes by at most .03 kappa points in either direction, with no consistent sign (two configurations decrease, one increases, one is essentially unchanged). Giving the model the same lookahead information available to the human raters therefore did not reliably improve alignment with the expert reference set. This indicates that the human–LLM disagreement reported in the main text is not primarily attributable to the blinded condition’s information asymmetry, and supports interpreting it instead as a genuine divergence in how models and human experts interpret the same evidence.

S6.4 Self-Consistency and Majority Vote

Each item was classified independently three times per model (self-consistency sampling), and the reported label for that item was the majority label across the three runs. Because three runs across six possible classes can produce a three-way split with no strict majority, ties were broken deterministically by first-seen run order; every record retains an explicit `unanimous` flag (true only if all three runs agreed) so that non-unanimous majority decisions remain distinguishable from unanimous ones in all downstream analysis. This self-consistency and majority-vote procedure was identical across all four model configurations in Table 5, including the tier-1 robustness check.

S7 Evaluation Metrics

All metrics were computed over the 100-item reference set unless otherwise noted, using the expert reference set (Section 4) as the human reference and each model’s majority-vote label (Section 6) as the corresponding prediction. The six-class label set {UNC, CTX, COMP, CON, PROC, NA} was used throughout; *Not Applicable* was retained as a regular label value in every computation below (never excluded), since whether a model appropriately abstains from diagnosis is itself part of what is being evaluated.

S7.1 Cohen’s κ

Pairwise agreement (each LLM vs. the expert reference set, and each pair of LLMs) was computed as Cohen’s κ (Cohen, 1960), implemented via scikit-learn’s `cohen_kappa_score`, over the fixed six-class label set.

S7.2 Krippendorff’s α

Multi-rater agreement across the three tier-2 LLMs jointly was computed as Krippendorff’s α (Hayes & Krippendorff, 2007), nominal level of measurement, via the `krippendorff` Python package, over the same six-class label set. Only the LLMs-only variant (excluding the human expert reference set) is reported in the main text.

S7.3 Bootstrap Confidence Intervals

Ninety-five percent confidence intervals for every κ and α point estimate were obtained via item-level bootstrap resampling: 1,000 resamples with replacement over the 100 reference-set items (random seed 42), recomputing the statistic on each resample, and taking the 2.5th and 97.5th percentiles of the resulting distribution.

S7.4 McNemar’s Test

Whether the expert reference set and each LLM differ systematically in how often they invoke the *Not Applicable* label was tested with the continuity-corrected McNemar test (McNemar, 1947), paired at the item level on the binary indicator “label = NA”:

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c}, \quad df = 1,$$

where b is the number of items the expert reference set labels NA but the LLM does not, and c is the number of items the LLM labels NA but the expert reference set does not (items on which both parties agree on NA-or-not-NA do not contribute to b or c). This test was run once per model, comparing that model against the expert reference set.

S7.5 Per-Class Precision, Recall, and F1

Per-class precision, recall, and F1 (expert reference set as reference, each LLM’s majority label as prediction) were computed via scikit-learn’s `precision_recall_fscore_support`. Two aggregations are reported in the main text: per-model per-class values, and the macro-average of per-class F1 across the three tier-2 models. A pooled confusion view additionally stacks all three tier-2 models’ predictions together (300 predictions over 100 items: each item contributes one prediction per model) to characterize, for each expert-reference-set class, which other class the models most often substitute for it. Classes with fewer than 10 expert-reference-set support items (Procedural Slip, $n = 6$; Operator Selection, $n = 6$) yield higher-variance per-class estimates and are interpreted accordingly in the main text.

S8 Ethics and Data Governance

The transcript analyses constitute exempt secondary analysis of de-identified, third-party tutoring data, as determined through [anonymized university]’s institutional review process; this determination was limited to secondary data analysis, and no additional human-subjects data were collected.

Research use of the corpus was confirmed directly with the data and tutoring provider prior to analysis. All identifying information (student names, tutor names, school identifiers) had already been removed by the data provider before the research team’s access, and no attempt was made to re-identify any individual. The speaker role-swap issue described in Section 1 was verified as a transcription/diarization defect in the source data rather than a defect introduced by this study’s processing pipeline and was reported to the data provider.

References

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